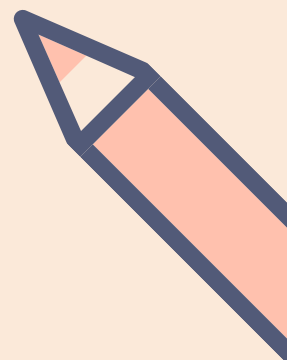
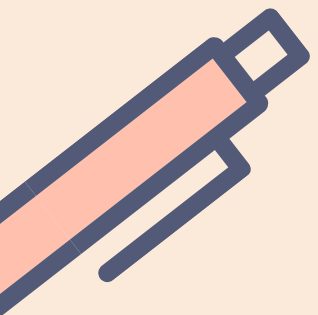




HOW TO CREATE A TNS SERVICE

using Net Configuration Assistant (NetCA)

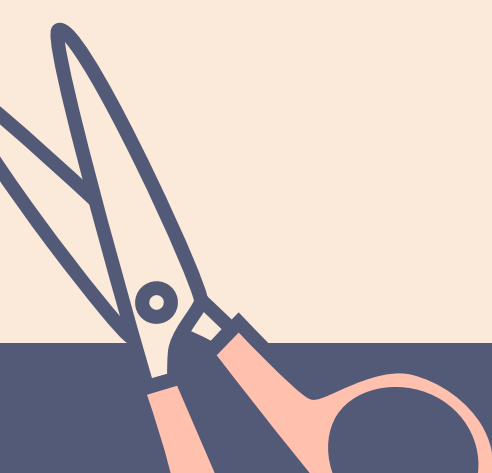


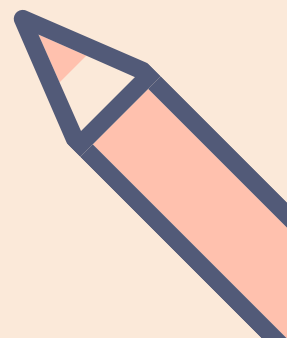
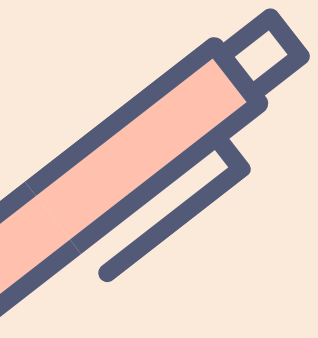
INTRODUCTION

what you need to create a TNS service

Thanks to Oracle Net Configuration Assistant, configuring a connect descriptor in tnsnames.ora file is a very easy task.

In order to configure a TNS entry you will need some information, such as –

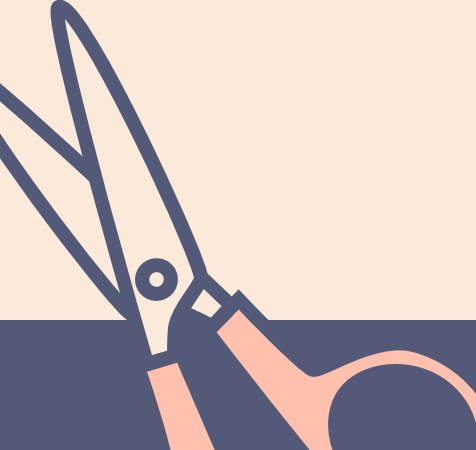
- **Service Name:** A service is a logical representation of a database. Every Database which you have configured on your server must have a corresponding service. You will need the name of the service of the database for which you are creating the entry in your tnsnames.ora file.
 - **Hostname:** Hostname will be the name of the machine where your Oracle Database Server is installed and configured.
 - **Port Number:** Port number is where the listener of your Oracle Database is listening or addressing the requests.
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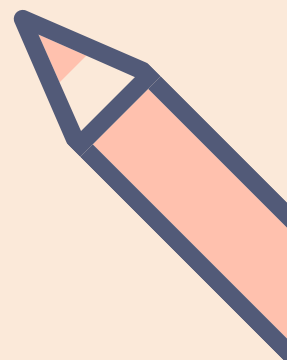
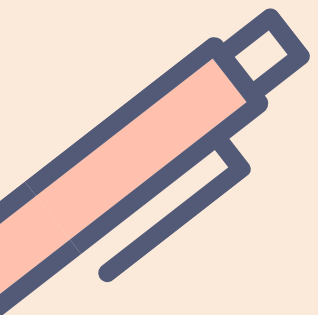


NetCA

Oracle Net Configuration Assistant is a wizard based graphical user interface (GUI). This tool comes bundled with Oracle Database. Using it you can configure almost all the Network settings of your Oracle Database Server.

Here I will show you how to use Oracle Net Assistant to create a TNS entry in tnsnames.ora file in an easy step-by-step manner.



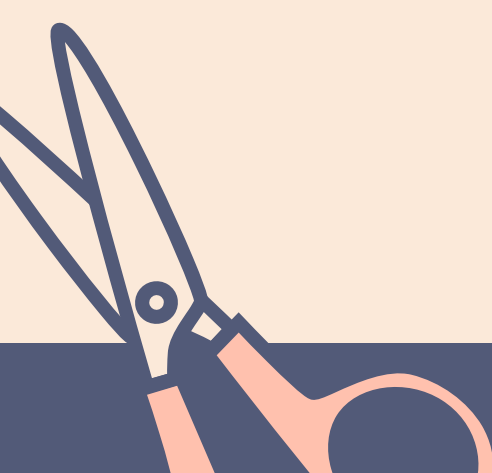


LAUNCH THE ORACLE NET CONFIGURATION ASSISTANT.

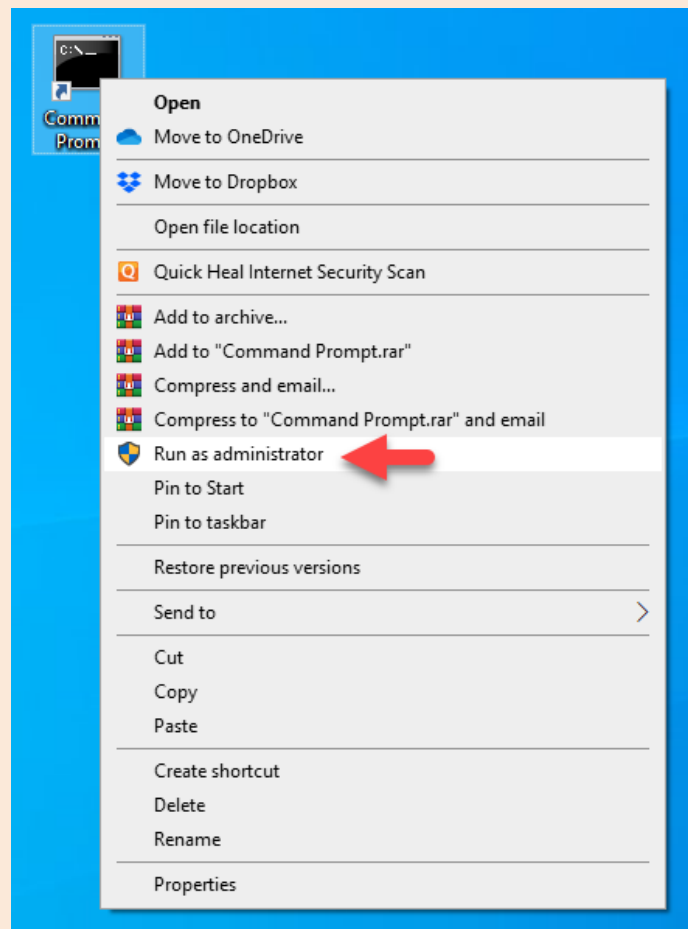
Have a productive vacation.

As Net Configuration Assistant is going to modify tnsnames.ora file by adding an entry into it thus it is important that you launch your NetCA with administrator privileges.

To launch the Oracle Net Configuration Assistant, we will use a Windows Dos/Linux command. That command is '**netca**'.



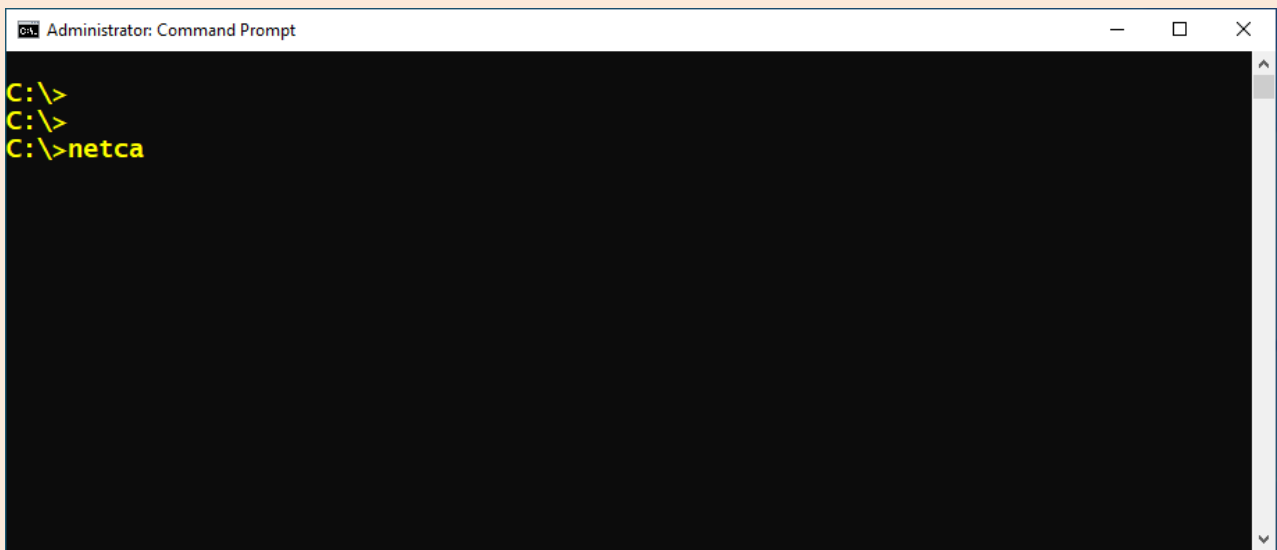
Run As Administrator



Run your command prompt with administrator privileges or shell with 'SUDO' rights.



The NetCA Command



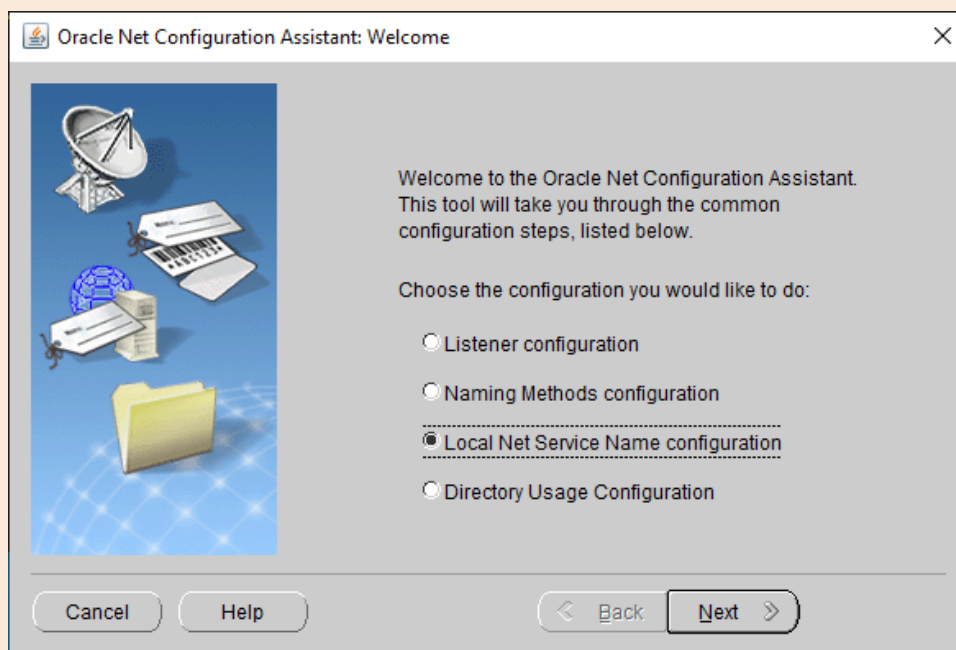
```
Administrator: Command Prompt
C:\>
C:\>
C:\>netca
```

After that simply write **netca** on the prompt. This will start the Oracle Net Configuration Assistant.



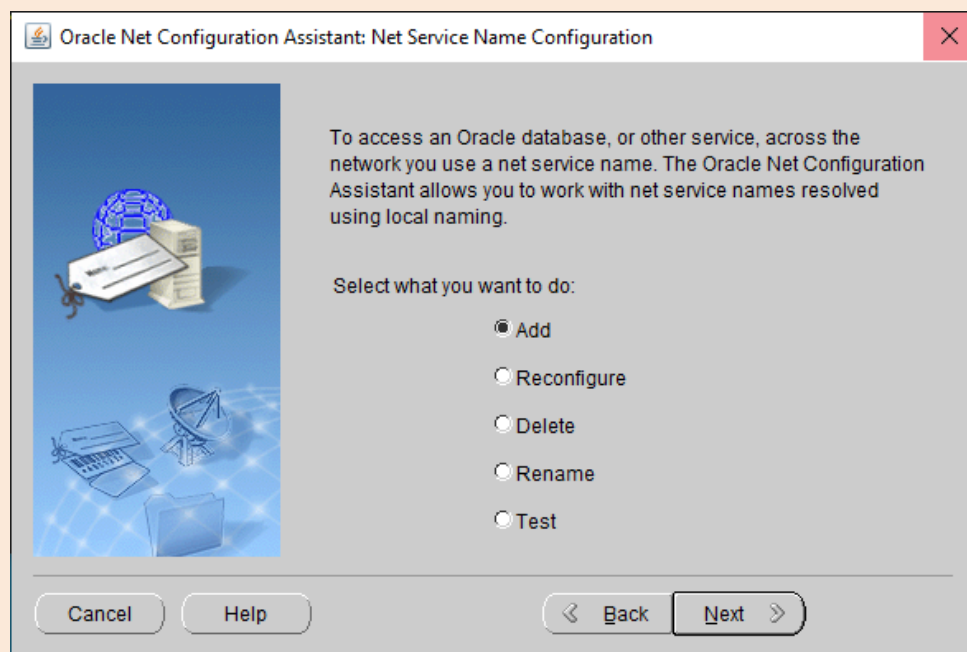
STEP 1: CHOOSE THE CONFIGURATION.

On the first screen of the net configuration assistant you will be provided with four options. To make a TNS entry in the tnsnames.ora file you have to choose the third option which is '**Oracle Net Service Name Configuration**'.



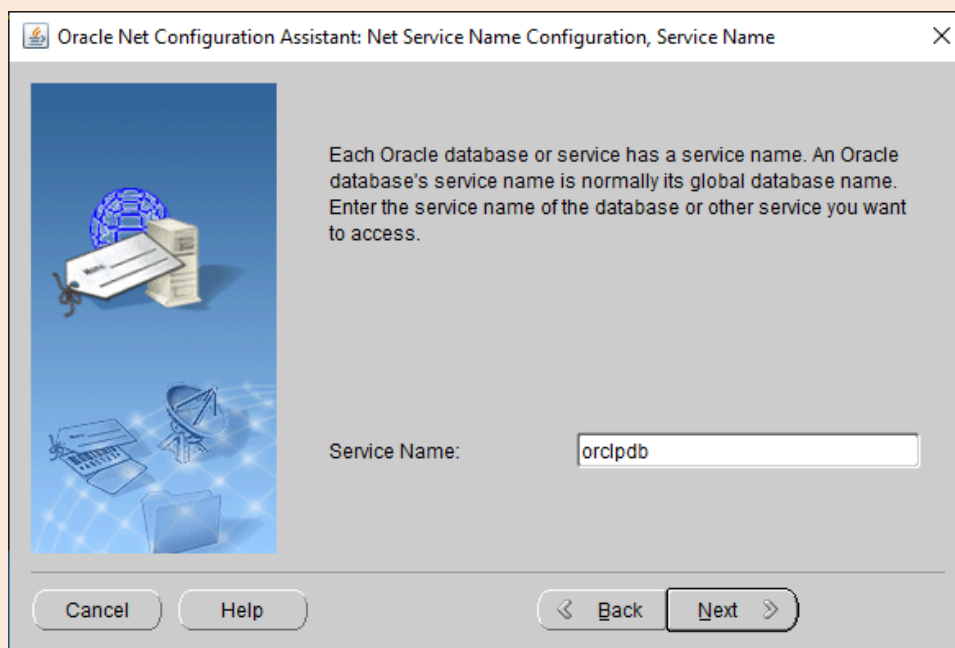
STEP 2: SELECT THE ACTION.

In this step you need to select the action from the list which you want to perform. As we want to add a TNS entry thus we will select the first option '**add**'.



STEP 3: THE SERVICE NAME

In the third step you will have to specify the service name for which you want to create this entry. Read this blog to learn how to find out the service name of a database.





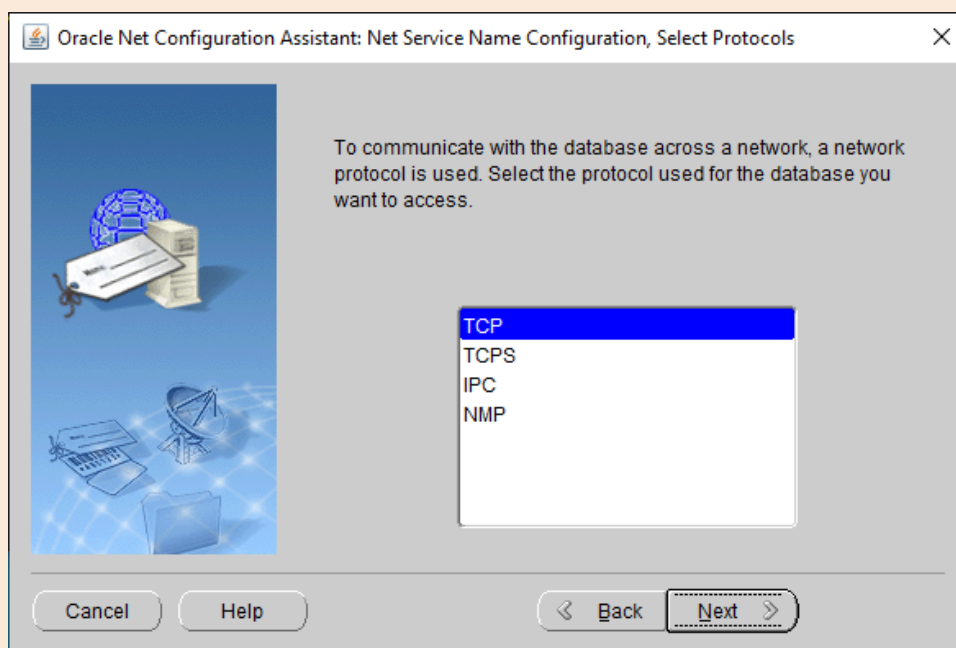
BEFORE THE NEXT STEP

In next three steps we will be configuring protocol address of the connect descriptor. The protocol address will be comprised of information like Network Protocol, hostname and the port number. You need to be very careful while specifying these information.

To find out the correct network protocol, hostname and the port number you can take help from the listener.ora file. As, in most cases the protocol address portion of the connect descriptor of the tnsnames.ora file is the same as that of the protocol address of the listener.ora file.

STEP 4: THE NETWORK PROTOCOL

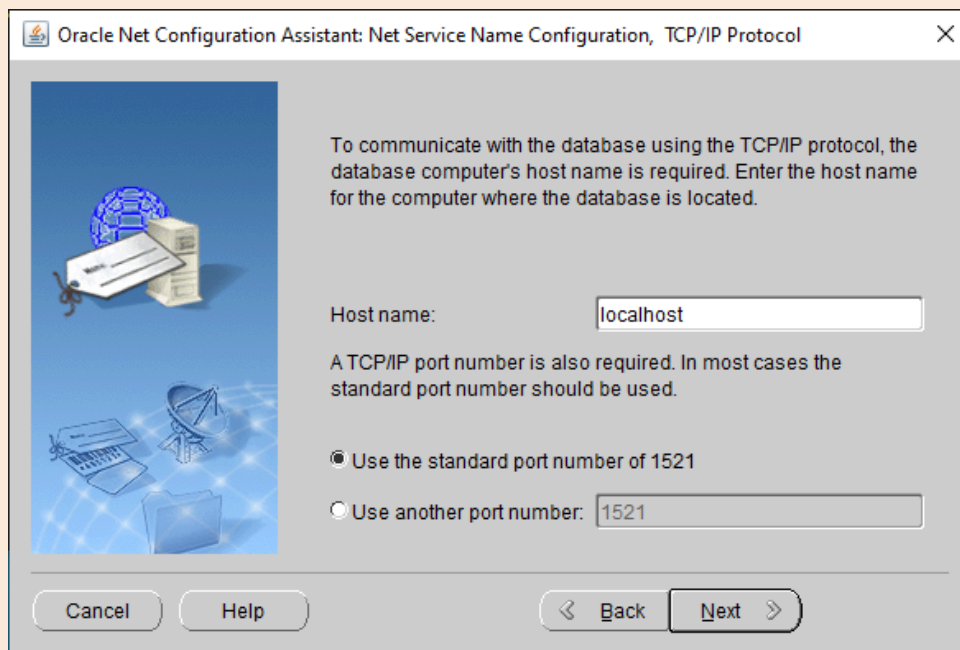
You need to select a protocol using which the client will communicate with your database on the network. In my machine the server-client communication is using TCP network protocol. Thus, here select **TCP** and move ahead.



STEP 5: THE HOSTNAME & THE PORT NUMBER

Here you have to specify **the name or the IP Address of the computer** where your Oracle Database server is installed and configured.

On this same screen you will also have to specify the port number which your Oracle Server is using for establishing the communication between Client and server. **By default, Oracle Server uses the port number 1521.**

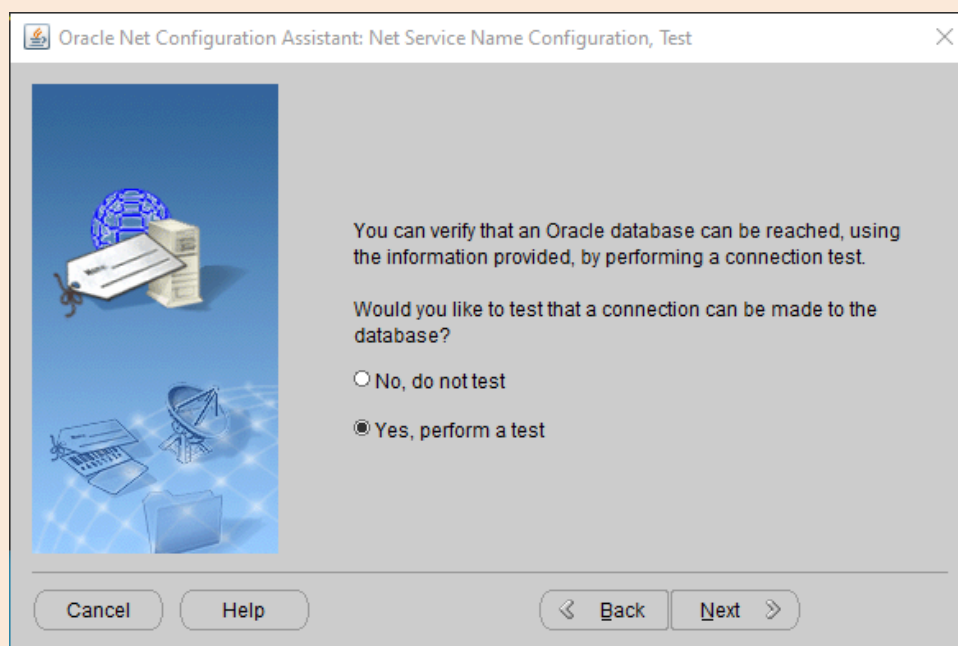


The screenshot shows a window titled "Oracle Net Configuration Assistant: Net Service Name Configuration, TCP/IP Protocol". On the left is a graphic with a globe, a computer tower, a keyboard, and a folder. The main text area contains the following instructions: "To communicate with the database using the TCP/IP protocol, the database computer's host name is required. Enter the host name for the computer where the database is located." Below this is a text box labeled "Host name:" containing the text "localhost". Further down, it says: "A TCP/IP port number is also required. In most cases the standard port number should be used." There are two radio button options: "Use the standard port number of 1521" (which is selected) and "Use another port number:" followed by a text box containing "1521". At the bottom are buttons for "Cancel", "Help", "< Back", and "Next >".

STEP 6: TEST THE TNS

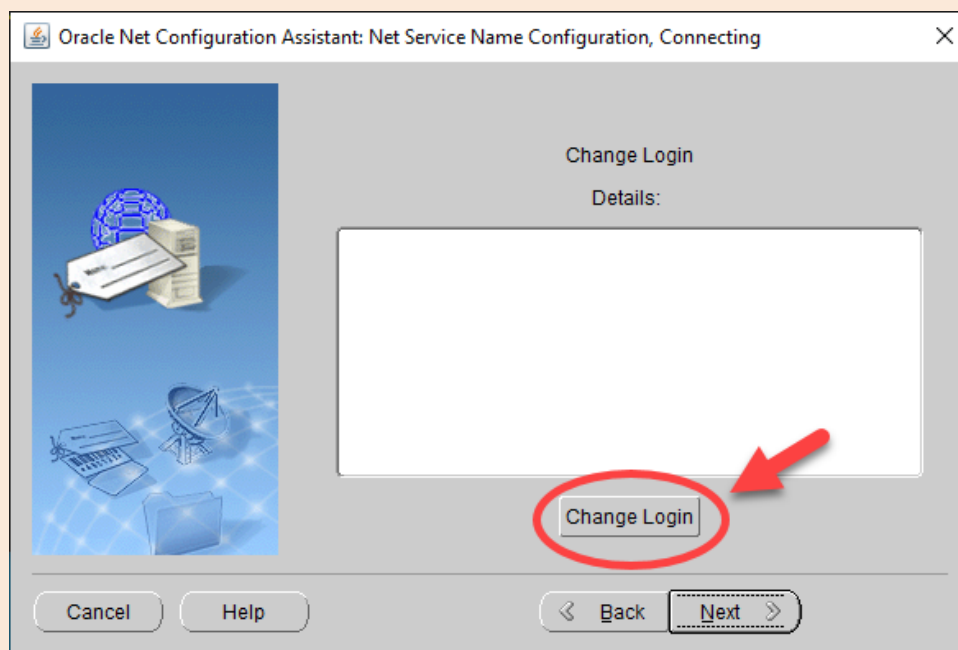
Once you are done creating your TNS entry, next you have to test it. Although it's not necessary but it is a good practice. Testing your TNS entry will let you know whether you have done the settings correctly or not.

Here Select '**Yes, perform a test**' and press the '**next**' button.



STEP 6: TEST THE TNS

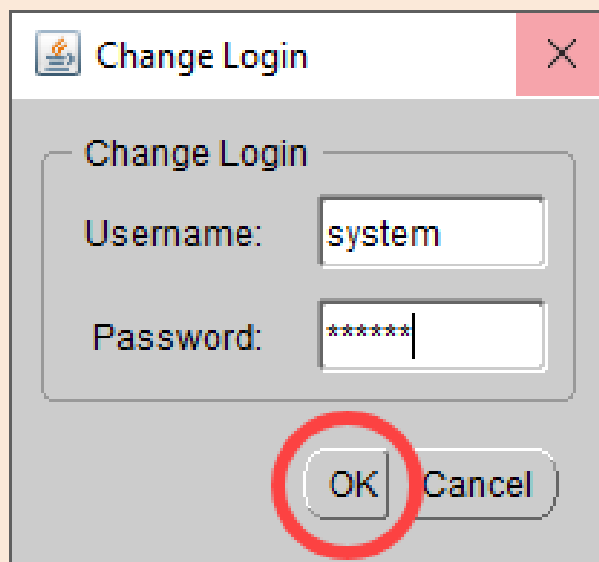
On the following screen press the change login button.



STEP 6: TEST THE TNS

Here enter '**system**' as **your username** and the password of your 'system' user. The password of your system user is the one which you specified during the installation of your Oracle Database.

After that just press **ok** and the Oracle Net Configuration Assistant will show you the result. If everything is ok, then you will get a success message otherwise the NetCA will show you the error.



Change Login

Change Login

Username: system

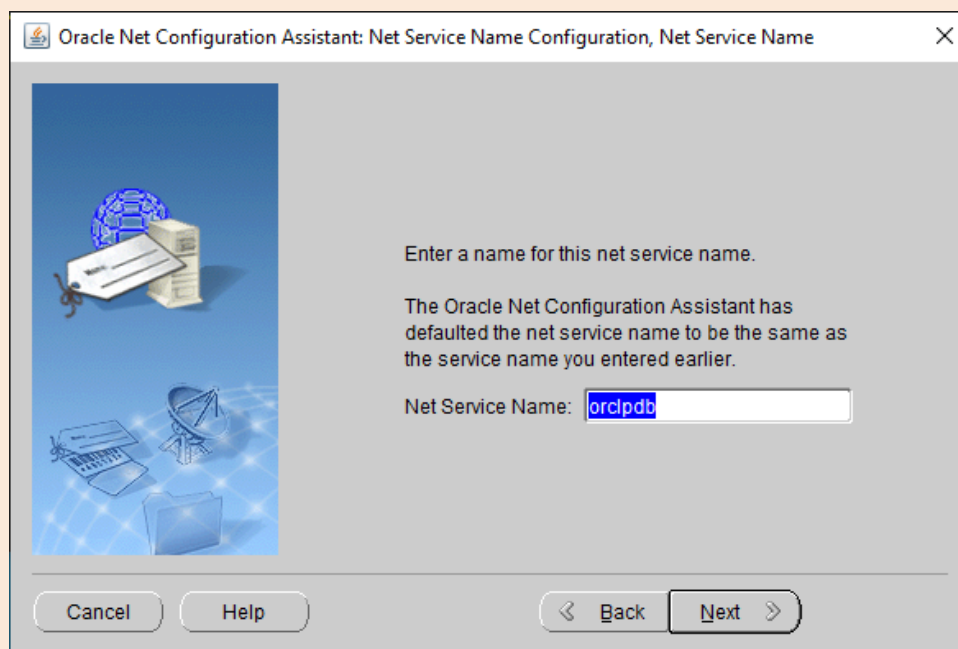
Password: *****

OK Cancel

STEP 7: THE SERVICE NAME

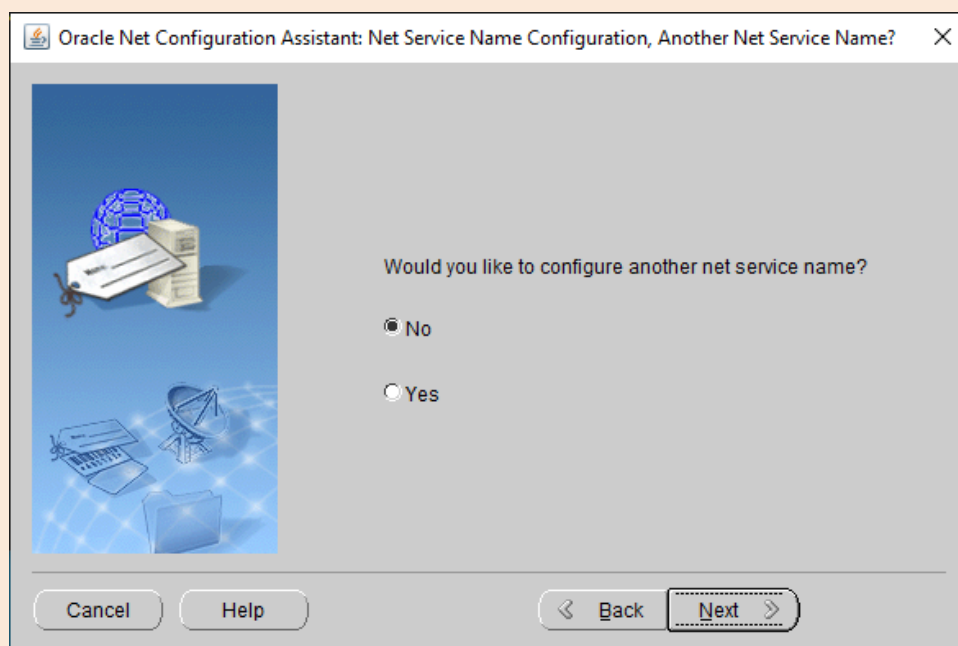
In this step you have to specify a name for your TNS entry. You can name your net service whatever you want. But it is always recommended to specify a logical name for your TNS service which could help you in recognizing it easily.

By default, the Oracle Net Configuration Assistant will name your TNS entry after the name of the service for which you are creating this connect descriptor. I suggest you not to change it.



STEP 8: ANOTHER NET SERVICE NAME

In this step you will be asked whether you want to configure another TNS entry or not. If you want to then select 'Yes' otherwise select 'No'.



STEP 9: NET CONFIGURATION DONE!

Congratulations you have created your connect descriptor/TNS service for your database.

